# Supplementary Material: Prompt Templates for QEG-RAG Two-Tier Diagnostic Engine

Trung-Kien Nguyen[1], Khac-Hoai-Nam Bui[2], Hoang-Hiep Lai[2], Quang-Huy Phung[2], and Van-Vinh Nguyen[1⋆]

[1] VNU University of Engineering and Technology, Vietnam National University, Hanoi, Vietnam
[2] Viettel AI, Viettel Group, Hanoi, Vietnam

## A Prompt Templates for Two-Tier Reasoning Engine

This supplementary document provides the exact prompt templates used in the Two-Tier Controlled Reasoning Engine of QEG-RAG.

### A.1 Tier 1: Primary Reader Prompt Template

The Tier 1 prompt serializes the compact evidence graph $\mathcal{X}^*(G_q)$, incorporating star record blocks, deduplicated context passages, and deterministic path scaffolds. It instructs the language model to perform immediate fact extraction or trigger a diagnostic handoff.

```
Answer the question using the grounded compact OUEG payload.

‘rows‘ retains table rows selected by grounded paths, their original order, values, passage
      references, and edge IDs. ‘edges‘ uses the legend schema and contains the other
     selected KG edges. ‘contexts‘ are deduplicated by passage ID. ‘paths‘ are reading
     guides only. Their ‘score‘ is rounded relevance metadata, not factual proof. Follow ‘
     ordering‘ when present; use ‘edges‘ and ‘contexts‘ for multi-hop evidence. Copy the
     shortest answer span exactly as written in the evidence; do not translate or expand
     names. Return only that span.

Fallback protocol (only when you cannot prove one answer from this payload): keep the
     normal successful-answer behaviour above unchanged---return its shortest answer span
     and nothing else. If you cannot answer, never return a blank response. Instead return
     exactly one of:
1. ‘NEED_FOCUS: {"candidates":["candidate names already in the payload"],"blocker_type":"
     missing_link","bottleneck":"why the answer is not yet unique","needed_check":"exact
     row, relation, year, rank, or passage fact to compare"}‘ when a focused L2 re-read can
      resolve it.
2. ‘UNRESOLVED: {"blocker_type":"insufficient_evidence","bottleneck":"what evidence is
     absent or contradictory","needed_check":"what fact would be required"}‘ when this
     payload cannot prove an answer.

‘blocker_type‘ must be exactly one label: ‘candidate_collision‘, ‘missing_link‘, ‘
     insufficient_evidence‘, ‘conflicting_evidence‘, or ‘answer_type_uncertain‘; never
     combine labels with ‘|‘.
Do not invent candidates or evidence.

Question: {question}
Compact OUEG payload: {compact_json}
Answer:
```

**Listing 1.1.** Tier 1 Prompt Template (Primary Reader with Diagnostic Protocol)

⋆ Corresponding author.

### A.2 Tier 2: Targeted Reader Prompt Template

When Tier 1 returns a `NEED_FOCUS` diagnostic signal, the targeted bridge retriever augments missing passages for disputed candidates without calling the LLM. Tier 2 then receives a minimal state capsule $\mathcal{K}_2$ containing *only* the disputed candidates and new bridge evidence, filtering out all unrelated table rows.

```
Resolve the question using only the grounded L2 state capsule below. This is a new request:
      do not assume you remember L1 or use facts absent from this capsule.

The capsule contains the original question, the unresolved constraint, complete row
     bindings for the focus candidates, and their relevant evidence. ‘coverage‘ says
     whether retrieval found new text; absence of evidence is not a negative fact. Scores
     are intentionally omitted.

Compare the candidates against the evidence to resolve the question. Return:
‘FINAL_ANSWER: <shortest evidence span>‘
‘EVIDENCE_PATH: {"edges":["edge IDs in this capsule"],"passages":["source IDs in this
     capsule"]}‘
for the proven candidate.

Only return ‘UNRESOLVED: {"blocker_type":"...","bottleneck":"...","needed_check":"..."}‘ if
      the evidence strictly contradicts all candidates. Never return blank or invent
     evidence.

Question: {question}
L2 state capsule: {compact_json}
Answer:
```

**Listing 1.2.** Tier 2 Prompt Template (Targeted Reader over Minimal State Capsule)